Marginal and Conditional Distributions KEY

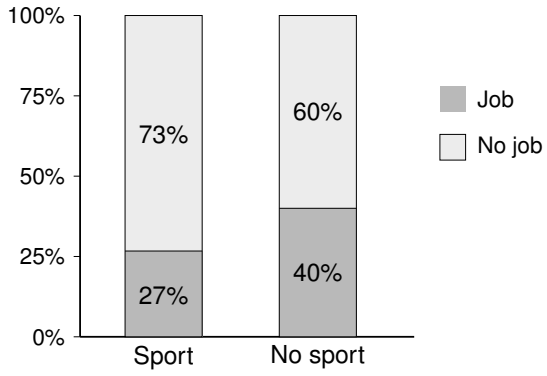
Name _____

Do students who play a sport have jobs as often as students who do not? To answer, compare percents **inside each group**.

	Job	No job	Total
Sport	8	22	30
No sport	12	18	30
Total	20	40	60

	What it is	Example
Marginal distribution	Percents from the **totals** (the margins). Divide by the **grand total**.	Job: $\frac{20}{60} \approx 33\%$ No job: $\frac{40}{60} \approx 67\%$
Conditional distribution	Percents inside **one group**. Divide by that group's **total**.	Students who play a sport: Job: $\frac{8}{30} \approx 27\%$ No job: $\frac{22}{30} \approx 73\%$

Compare the conditional distributions.



Sport: $\frac{8}{30} \approx 27\%$ have a job.

No sport: $\frac{12}{30} = 40\%$ have a job.

The percents are **different**. Playing a sport and having a job are **associated**. In this group, students who play a sport are less likely to have a job.

If the percents were about the **same**, there would be **no association**.

Gotcha: check the denominator. "Of the students who **play a sport**, what percent have a job?" Divide by the sport total (30), not by 60 and not by the job total (20).

Work with a partner. Compare answers when you finish each problem.

1. Use the table of how 80 students get to school.

	Bus	**Car**	**Walk**	**Total**
9th grade	18	12	10	40
10th grade	14	20	6	40
Total	32	32	16	80

   Find the marginal distribution of how students get to school.

   Bus: $\frac{32}{80} = 40\%$. Car: $\frac{32}{80} = 40\%$. Walk: $\frac{16}{80} = 20\%$.

2. Use the same table.

   (a) Find the conditional distribution of how 9th graders get to school.

   Bus $\frac{18}{40} = 45\%$, Car $\frac{12}{40} = 30\%$, Walk $\frac{10}{40} = 25\%$

   (b) Find the conditional distribution of how 10th graders get to school.

   Bus $\frac{14}{40} = 35\%$, Car $\frac{20}{40} = 50\%$, Walk $\frac{6}{40} = 15\%$

   (c) Is grade associated with how students get to school? Explain.

   Yes. The percents are different. For example, 50% of 10th graders come by car, but only 30% of 9th graders.

3. Use the same table.

   (a) What percent of 10th graders come by car? $\frac{20}{40} = 50\%$

   (b) What percent of car riders are 10th graders? $\frac{20}{32} = 62.5\%$

   (c) Why are the answers to (a) and (b) different?

   They divide by different groups: (a) by the 40 10th graders, (b) by the 32 car riders.

4. Is each statement about the school table a **marginal** or a **conditional** percent?

   (a) 40% of all students ride the bus. Marginal

   (b) Of the students who walk, 62.5% are 9th graders. Conditional (only walkers)

   (c) Half of the students are 10th graders. Marginal

   (d) 25% of 9th graders walk. Conditional (only 9th graders)

5. 100 students said if they wear glasses and if they play an instrument.

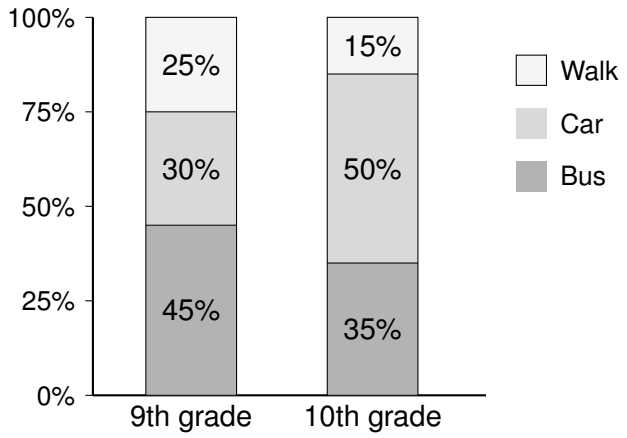
	Glasses	**No glasses**	**Total**
Instrument	12	28	40
No instrument	18	42	60
Total	30	70	100

   (a) What percent of students who play an instrument wear glasses? $\frac{12}{40} = 30\%$

   (b) What percent of students who do not play an instrument wear glasses? $\frac{18}{60} = 30\%$

   (c) Is playing an instrument associated with wearing glasses? Explain.
   No. The percents are the same (30%) in both groups.

6. This segmented bar chart shows the conditional distributions from problem 2.



   (a) Which grade has the larger percent of students who walk? 9th grade (25% vs 15%)

   (b) How does the chart show that grade and transportation are associated?
   The two bars are split differently. If there were no association, the bars would look about the same.